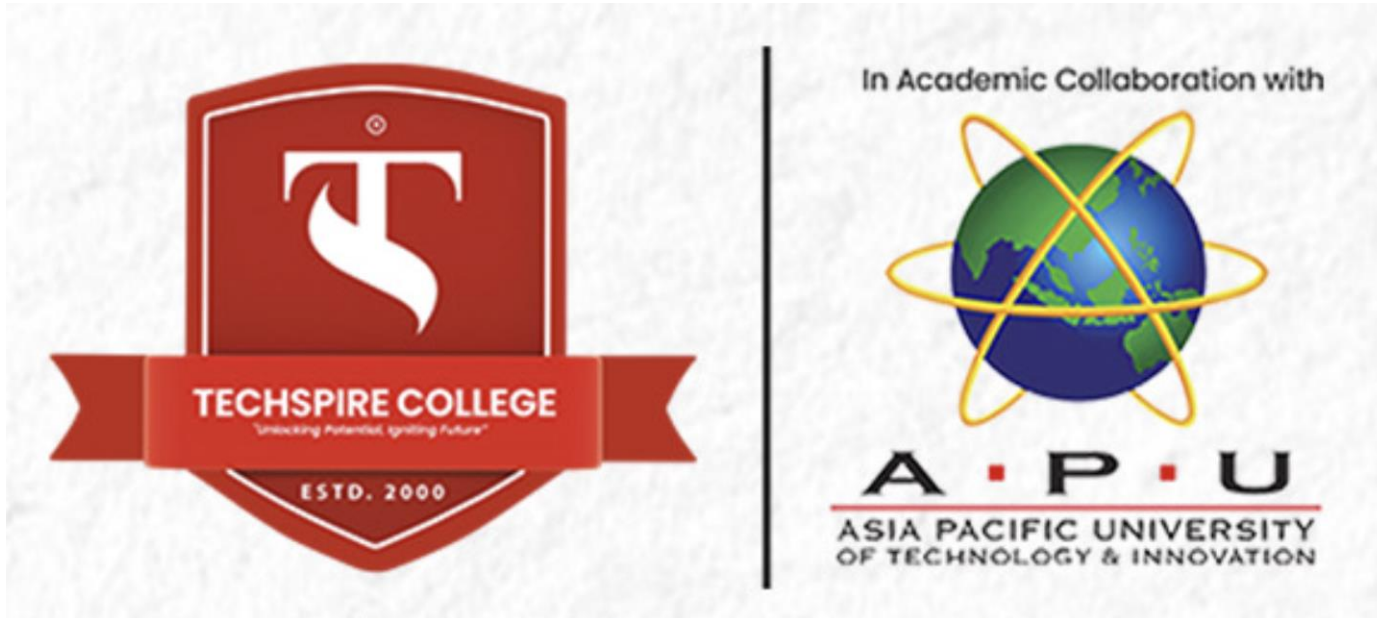




OOP in Java

Slide 5 - Arrays – Storing Multiple Values

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What Are Arrays?

- An array is a collection of multiple values stored in a single variable. All values in an array are of the same type (e.g., integers, strings)
- Why Use Arrays?
 - Imagine you have a row of lockers, each with a number. Instead of carrying 10 separate bags, you store them in those lockers. An array is like a row of lockers, where each locker can store a value.

Array in Java

- An **array** is a data structure that holds a fixed number of values of the same type.
- Basic Syntax
 - `dataType[] arrayName = new dataType[size];`
 - `dataType`: The type of values the array will hold (e.g., `int`, `String`).
 - `arrayName`: The name of the array (you can choose this).
 - `size`: The number of elements the array will hold.



Declaring and Initializing an Array

```
// Declare an array to hold 5 integers.  
int[] numbers = new int[5];  
// Explanation: This creates an array  
// called numbers that can store 5 integers.
```

```
// Populating the Array:  
// You can assign values to each element in the array.  
numbers[0] = 10;  
numbers[1] = 20;  
numbers[2] = 30;  
numbers[3] = 40;  
numbers[4] = 50;  
// Explanation: The array has 5 slots  
// (from numbers[0] to numbers[4]). Each slot can hold a value.
```

Accessing Array Elements

- Print the first element of the array.
 - `System.out.println(numbers[0]); //`
Output: 10
 - To access an element in the array, use the **index** (position) of the element. The first element is at index 0, the second at 1, and so on.

```
// Example: Use a for loop to print all elements in the array.  
for (int i = 0; i < numbers.length; i++) {  
    System.out.println(numbers[i]);  
}
```

Using a Loop to Access All Elements

- `numbers.length` gives the size of the array (5 in this case).
- The loop starts at `i = 0` and runs until `i < numbers.length`.
- Inside the loop, `numbers[i]` prints each element.

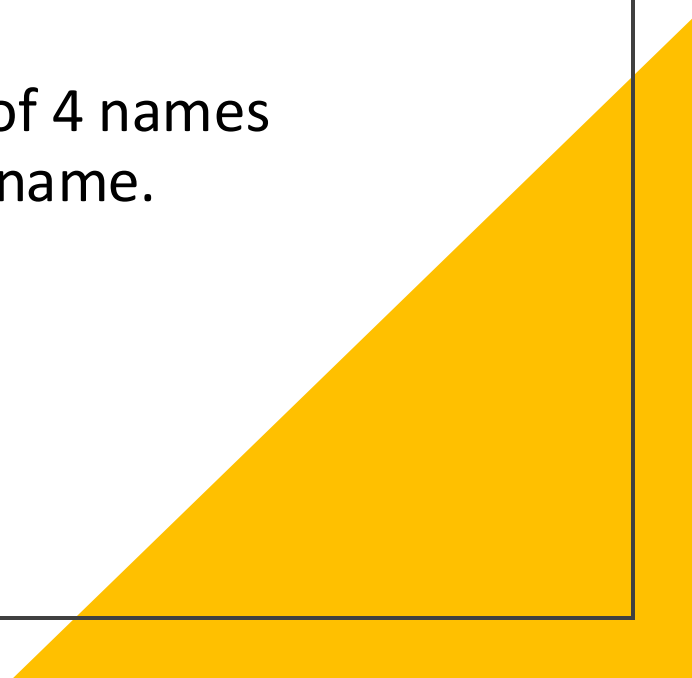
Declaring and Initializing Arrays in One Step

- Declare and initialize an array of 3 names in one step.
 - `String[] names = {"Alice", "Bob", "Charlie"};`
 - You can declare and fill an array at the same time by listing the values inside {}.

Changing Array Elements

- Change the second element in the names array.
 - `names[1] = "David"; // Replaces "Bob" with "David"`
 - Arrays are **mutable**, which means you can change the values of elements after the array is created.

Exercise

- **Task 1:** Create an array to store the first 5 positive numbers and print them.
 - **Task 2:** Create an array of 4 names and change the second name.
- 
- A large yellow triangle is positioned in the bottom right corner of the slide, pointing towards the top right.

Summary

Array: A collection of values of the same type stored together.

Accessing Elements: Use the index (starting from 0) to access array elements.

Loops and Arrays: for loops are commonly used to go through each element in an array.